We used a dataset with 1,500 varying prompts from the *Alpaca Dataset*. For each trait on our list, we presented the LLM with both the neutral prompt from the Alpaca Dataset and the trait-specific prompt. We next recorded the activations from Layer 18, which we have previously identified as the most influential layer for personality expression, for both neutral and trait-elicited responses. These activation vectors were saved for later study, and they provided the raw data for generating the personality direction vectors as mentioned in the previous section.

2.4 Structure of the Personality Space

After establishing our method to extract personality-specific activations, we turned our attention to examining the structure and connections within the resulting personality vector space. This investigation aims to reveal potential underlying dimensions of personality representation within the LLM, shedding light on how the model categorizes and differentiates various personality traits.

2.4.1 Visualizing Personality Activations

Before exploring the global structure of the personality vector space using dimensionality reduction techniques, we first visualize the activation patterns associated with specific personality traits across different layers of our LLM.

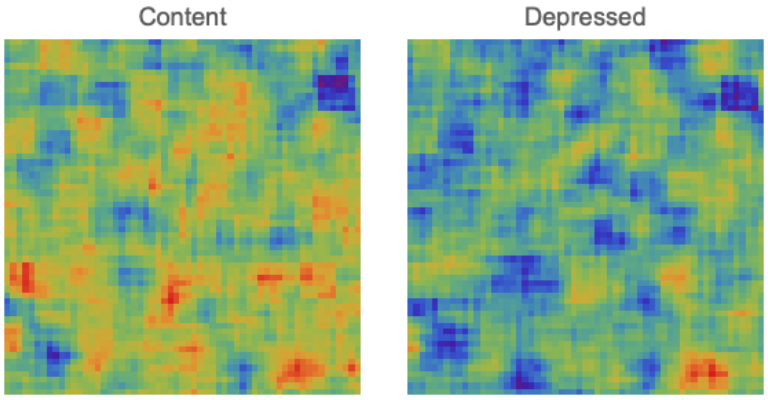

Figure 1: Activation patterns in Layer 18 for "Content" and "Depressed" personalities. Each pixel represents a group of dimensions from the 4,096-dimensional activation vector. Brighter colors, particularly towards red, indicate larger changes in activation compared to the baseline.

Similar to how neuroimaging studies have shown that specific emotions can be decoded from distinct patterns of brain activation across multiple regions[5, 9, 11], we can observe how the LLM's layers respond to personality activations. Figure 1 shows an example of these activations for the "Content" and "Depressed" personalities in Layer 18 of our Llama 3 8B model (which has 31 layers in total).

While the x and y axes in the visualization do not correspond to specific features, they organize the 4,096 dimensions of the activation vector into a 2D image for easier interpretation. The colors indicate the magnitude of change in activations compared to the model's baseline behavior without any personality prompt.

Interpreting the Visualizations Brighter or more intense colors, especially those towards the red end of the spectrum, highlight areas where the personality prompt induced significant changes in the model's internal state. Similarities in activation patterns between the two personalities might suggest regions involved in general "personality encoding," while differences could point to trait-specific adaptations.

Interestingly, our initial observations suggest that the most prominent differences in activations tend to occur in the middle layers of the model. This suggests that higher-level representations of personality might emerge in these intermediate layers.

2.4.2 Dimensionality Reduction

The activation vectors, sourced from Layer 18 (which we previously found to be highly responsive to personality adjustments), exist in a high-dimensional space. To facilitate visualization and exploratory analysis, we employed three dimension reduction techniques: Principal Component Analysis (PCA), t-Distributed Stochastic Neighbor Embedding (t-SNE), and Uniform Manifold Approximation and Projection (UMAP). These methods allowed us to project the personality vectors onto a more manageable two-dimensional plane.

Figure 2 presents the outcomes of these visualizations. Although each technique highlights different aspects of the data's structure, some consistent patterns emerge. Notably, traits with similar meanings, such as "introverted" and "solitary," tend to group together. This clustering suggests that the LLM captures meaningful semantic relationships between personality attributes.

2.4.3 Clustering for Structural Analysis

To gain a clearer picture of the underlying structure, we applied clustering analysis to the original high-dimensional personality vectors. We used the K-means algorithm to sort the traits into 20 distinct groups based on their closeness within the activation space. This method helps us identify sets of personality traits that the LLM represents in similar ways, offering a more detailed view of its internal personality framework.

Figure 3 shows the clustering results mapped onto a two-dimensional space using t-SNE. The proximity of points indicates similarity between traits, while the colors highlight distinct clusters. *Table 3* provides a full breakdown of each cluster and its associated personality traits.

The clustering results offer compelling insights into how the LLM organizes personality concepts. A notable example is *Cluster 8*, which combines various negative and potentially harmful traits.

Figure 2: *Two-dimensional visualization of 100 randomly sampled personality vectors using (a) PCA, (b) UMAP, and (c) t-SNE.*